

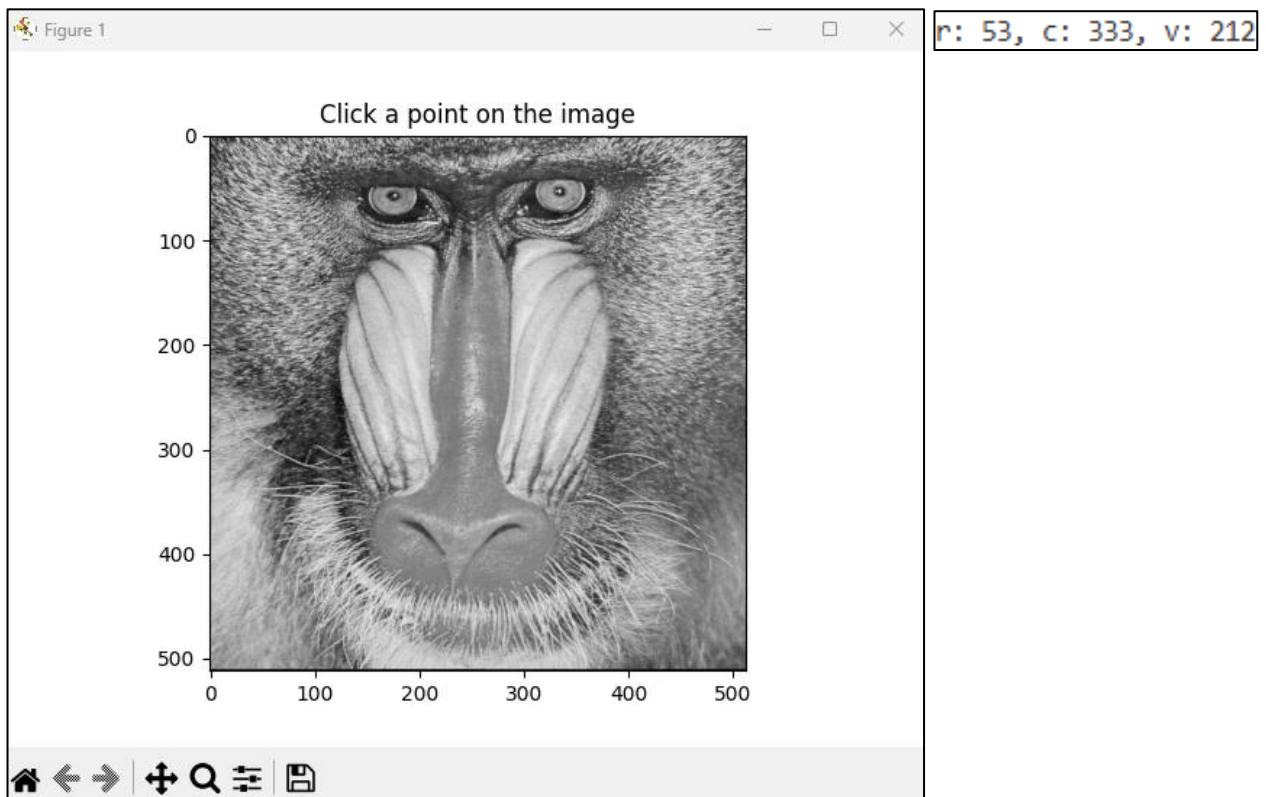
Exercise for Assignment 1 (Week 3)

Extracting pixel values from grayscale image:

- (a) Write a function *pixVal4e*(*f*, *r*, *c*), where *f* is a grayscale image and *r* and *c* are scalars corresponding respectively to a row and column number in *f*. Output *v* is the pixel value *f*(*r*, *c*).
- (b) Test your function by reading the image “BaboonGray.jpg” and obtaining the pixel values at the origin and at the middle of the image. Display the values on the screen.

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Origin: 146  
Image Center: 192
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- (c) Write a function *cursorValues4e*(*f*) that displays image *f*, displays a mouse-controlled cursor over it and, when the mouse left button is clicked, outputs the row/column coordinates (*r*, *c*) and the value *v* of the pixel at those coordinates. (*Hint*: Use the function from (a) to obtain the pixel value and use function **plt.ginput** to display to get the cursor’s coordinates)
- (d) Read the image and use *cursorValues4e* to obtain the coordinates and value of the center of the baboon’s right pupil. Display the results.



Requirements:

- Only the following packages are permitted: OpenCV-Python, Matplotlib
- Use Python version 3.10+